

Word family: **solve / solved / solving / solution / solvable / unsolvable / solver / problem-solving**

Base word: solve

Verb forms: solve, solves, solved, solving

Related forms: solution, solvable, unsolvable, solver

Useful extensions: problem-solving, solution-oriented

Part of speech: verb (動詞) / noun (相關名詞) / adjective (相關形容詞)

Your sentence:

> “Overall, that advertisement was persuasive because it **solved a real problem** and made the product seem useful, comforting, and luxurious.”

A full family sweep shows that **solve** has one very strong core meaning rather than many unrelated meanings:

> **to find an answer to, or successfully remove, a difficulty/problem**

The different uses—**solve a problem, solve a mystery, solve an equation, solve a crime**—all grow from this same core idea. So I would keep them connected rather than artificially splitting them into completely different meanings.

Meaning in this passage

Here, **solved a real problem** means:

the product provided an effective practical answer to a genuine difficulty the speaker experienced—noise during commuting/travel.

繁體中文: **有效處理並解決了一個真實存在、會影響日常生活的實際問題**

Natural translation:

> “Overall, that advertisement was persuasive because it solved a real problem...”

「整體而言，那則廣告很有說服力，因為它展示的產品**確實能解決一個現實生活中的實際問題**……」

Here:

solve a problem = 找到並採取有效方法，使一個困難或問題不再存在／得到處理

1. **solve + problem/difficulty (問題／困難) — 解決; 處理**

This is your original meaning.

****Meaning:**** To find an effective answer or way of dealing with a problem so that it is no longer a problem.

Practice 1 — Original

> “The headphones ****solved a real problem****.”

「這副耳機****解決了一個真實存在的實際問題****。」

Practice 2

> “Better insulation could ****solve the noise problem****.”

「更好的隔音措施可能可以****解決噪音問題****。」

Practice 3

> “We need to ****solve this problem**** before it gets worse.”

「我們需要在問題惡化之前把它****解決****。」

****solve a problem = 找到並實施有效方法, 使困難得到處理或消除****

2. ****solve + puzzle/riddle(謎題／謎語) — 解開; 找出答案****

****Meaning:**** To discover the correct answer to something designed to be difficult or puzzling.

Practice 4

> “Can you ****solve this puzzle****?”

「你能****解開這道謎題****嗎？」

Practice 5

> “It took me twenty minutes to ****solve the riddle****.”

「我花了二十分鐘才****猜出／解開那個謎語****。」

****solve a puzzle/riddle = 經過思考找出謎題或謎語的正確答案****

This is the same core concept: **difficulty** → **correct answer**.

3. **solve + equation/mathematical problem (方程式／數學題)** — **解答; 求解**

A major academic use.

Practice 6

> “Can you **solve this equation**?”

「你能**解這條方程式**嗎？」

Practice 7

> “The students learned how to **solve quadratic equations**.”

「學生學習如何**解二次方程式**。」

solve an equation = 透過數學步驟找出符合方程式條件的未知數值

4. **solve for + variable (未知數)** — **求.....的值**

A useful mathematical pattern.

Practice 8

> “**Solve for x**.”

「**求 x 的值**。」

Practice 9

> “First, rearrange the equation and then **solve for y**.”

「首先重新整理方程式, 然後**求出 y 的值**。」

solve for + variable = 計算並找出指定未知數的值

5. ****solve + mystery (謎團／疑問) — 解開; 查明真相****

****Meaning:**** To discover the explanation behind something mysterious or unexplained.

Practice 10

> “Scientists are trying to ****solve the mystery****.”

「科學家正嘗試****解開這個謎團／找出真相****。」

Practice 11

> “The final clue helped her ****solve the mystery****.”

「最後一條線索幫助她****解開了謎團****。」

****solve a mystery = 找出一件原本不明或神秘事情背後的真正解釋****

6. ****solve + crime/case (罪案／案件) — 破案; 查明案件真相****

A common police/legal use.

****Meaning:**** To discover who committed a crime and establish what happened.

Practice 12

> “Police finally ****solved the case****.”

「警方終於****破了案／查明案件真相****。」

Practice 13

> “New evidence helped detectives ****solve the crime****.”

「新證據幫助偵探****偵破了這宗罪案****。」

****solve a crime/case = 找出犯案者及案件經過, 使案件得到偵破****

Again, same underlying concept:

> ****unknown answer → answer discovered****

7. **solve + technical/scientific issue** (技術／科學問題) — 解決; 找出解決方法

Practice 14

> “Engineers are working to **solve the battery problem**.”

「工程師正努力**解決電池方面的問題**。」

Practice 15

> “The update **solved several technical issues**.”

「這次更新**解決了幾個技術問題**。」

solve a technical issue = 找出並實施方法, 使技術故障或限制得到消除

This is still the general **problem-solving** sense.

8. **solution = answer to a problem** (問題／困難) — 解決方法; 解決方案

This is the most important noun in the family.

Practice 16

> “Noise-cancelling headphones were a practical **solution** to the problem.”

「降噪耳機是這個問題的一個實際**解決方案**。」

Practice 17

> “We need to find a long-term **solution**.”

「我們需要找到一個長遠的**解決方法**。」

solution = 能夠有效處理、減少或消除某個問題的方法／答案

So:

> **solve** = 解決(動作)

> **solution** = 解決方法／方案(名詞)

9. **solution** = correct answer (題目／謎題／數學) — 解答; 答案

Practice 18

> “The teacher explained the **solution** to the equation.”

「老師解釋了這條方程式的**解答**。」

Practice 19

> “There may be more than one possible **solution**.”

「這道題可能有不止一個可行的**解答**。」

solution = 經過推理或計算後所得的正確答案

This is the noun counterpart of **solve a puzzle/equation**.

10. **solution** = product/service offered to meet a need (商業／科技) — 解決方案; 整套服務／系統

Very common in business and technology.

Practice 20

> “The company provides cloud-based security **solutions**.”

「公司提供雲端保安**解決方案**。」

Practice 21

> “We need a more flexible software **solution**.”

「我們需要一套更靈活的軟件**解決方案**。」

solution — business/technology = 為解決某種客戶需求或問題而設計的一套產品、服務或系統

This can sometimes sound like corporate jargon, but it is very common.

11. **solution(化學) — 溶液**

This is a genuinely distinct meaning of **solution** and should not be missed in the family sweep.

Meaning: A liquid mixture in which one substance has dissolved evenly in another.

Practice 22

> “Salt dissolves in water to form a **solution**.”

「鹽溶於水後形成**溶液**。」

Practice 23

> “The scientist added the chemical **solution** slowly.”

「科學家慢慢加入那種化學**溶液**。」

solution — chemistry = 一種物質均勻溶解於另一種物質後形成的混合物

Important: this does **not** mean 「解決方案」 in chemistry.

12. **solvable = capable of being solved(問題／題目) — 可以解決的; 有解的**

Practice 24

> “The problem is difficult but **solvable**.”

「這個問題雖然困難, 但**是可以解決的**。」

Practice 25

> “Not every puzzle is easily **solvable**.”

「並不是每個謎題都**容易解開**。」

solvable = 存在可行方法或答案, 可以被成功解決／解答的

13. **unsolvable = impossible or extremely difficult to solve — 無法解決的; 無解的**

Practice 26

> “At first, the problem seemed **unsolvable**.”

「起初, 這個問題看起來**無法解決**。」

Practice 27

> “The mystery remained **unsolvable** for decades.”

「這個謎團數十年來一直**無法破解**。」

unsolvable = 沒有可行解決方法／答案, 或以現有條件無法成功解決的

14. **solver** = person/system that solves something — 解題者; 求解工具／求解器

Practice 28

> “She is an excellent problem **solver**.”

「她是一個很出色的**問題解決者**／很擅長解決問題的人。」

Practice 29

> “The software includes an equation **solver**.”

「這套軟件包含一個方程式**求解器**。」

solver = 能夠找出問題、謎題或方程式答案的人、程式或工具

15. **problem-solving** = process/skill of finding solutions — 解決問題; 問題解決能力

A very important family extension.

Practice 30

> “The job requires strong **problem-solving skills**.”

「這份工作需要很強的**解決問題能力**。」

Practice 31

> “The exercise develops logical **problem-solving**.”

「這項練習可以培養邏輯性的**問題解決能力**。」

problem-solving = 分析問題、找出原因、比較可能方法並找到有效解決方案的過程／能力

16. **solution-oriented** = focused on finding ways to solve problems — 以解決問題為導向的

A useful modern workplace extension.

Practice 32

> “She takes a **solution-oriented** approach.”

「她採取一種**以尋找解決方法為導向的做法**。」

Practice 33

> “Good managers should be practical and **solution-oriented**.”

「好的經理應該務實，而且**著重尋找解決方案**。」

solution-oriented = 遇到問題時主要著眼於找出可行解決方法，而不是只停留在問題本身

Important distinction: **solve vs fix**

These overlap, but not completely.

solve

usually focuses on a **problem, puzzle, mystery, equation, issue**

> “solve a problem”

fix

often focuses on repairing something broken or correcting a fault

> “fix a broken laptop”

「修理壞掉的手提電腦」

Compare:

> “We solved the problem.”

「我們解決了問題。」

> “We fixed the computer.”

「我們修好了電腦。」

You can **fix a problem** too, especially conversationally, but **solve a problem** is the more general standard collocation.

Important distinction: **solve** vs **resolve**

These are close but not identical.

solve

finds an answer to a problem.

> “solve a technical problem”

resolve

often means:

- * settle a disagreement;
- * bring an issue to a satisfactory conclusion;
- * make a firm decision.

> “resolve a dispute”

「解決／平息爭議」

> “resolve an issue”

「處理並解決問題」

So **solve** is especially natural when the problem has an answer or practical solution.

Important distinction: **solve** vs **answer**

answer

respond to a question.

> “answer a question”

solve

work out a problem.

> “solve a problem”

So normally:

> **answer the question** ✓

> **solve the problem** ✓

For mathematical questions, you may **solve the problem** and then **give the answer**.

Important distinction: **solve** vs **deal with**

solve

implies the problem has actually been successfully dealt with.

deal with

means take action concerning it, but does **not necessarily mean it disappears**.

> “We are dealing with the problem.”

「我們正在處理這個問題。」

The problem may still exist.

> “We solved the problem.”

「我們把問題解決了。」

The outcome is much stronger.

This distinction is especially important in your original sentence.

The headphones did not merely:

> **deal with noise**

They **solved a real problem** for the speaker:

> the noise was no longer interfering so much with travelling/concentration.

Important distinction: **solution vs answer**

answer

usually gives the correct response to a question.

solution

often includes the method or outcome that resolves a problem.

> “The answer is 12.”

「答案是 12。」

> “Here is the solution to the problem.”

「這是這道題的解法／解答。」

In mathematics, **solution** can also mean a value satisfying an equation.

Important distinction: **solution vs resolution**

solution

= method/answer that solves a problem

resolution

can mean:

- * successful settlement of an issue;
- * formal decision;
- * image detail/clarity.

> “a solution to the problem”

「問題的解決方案」

> “resolution of the dispute”

「爭議的解決」

They overlap only in some contexts.

Important grammar: **solution to, not normally solution of**

The standard pattern for a problem is:

> **the solution to the problem** ✓

> **a solution to traffic congestion** ✓

Not normally:

> *the solution of the problem* ✗ in ordinary English

However, **solution of an equation/problem** may occur in formal mathematics to mean the process/result of solving it.

For normal everyday English, remember:

> **solution to + problem**

Important collocation note

We usually say:

- * **solve a problem**
- * **solve a mystery**
- * **solve a puzzle**
- * **solve an equation**
- * **solve a crime/case**

But:

- * **answer a question**
- * **fix a machine**
- * **settle/resolve a dispute**

Choosing the natural noun after **solve** matters.

Family-extension sweep

After the full family sweep, the useful modern family includes:

- * **solve** = 解決／解答／破解
- * **solve a problem** = 解決問題
- * **solve an equation** = 解方程式
- * **solve for x** = 求 x 的值
- * **solve a mystery** = 解開謎團
- * **solve a crime/case** = 破案
- * **solution — problem** = 解決方法／方案
- * **solution — answer** = 解答
- * **solution — business/tech** = 解決方案／整套產品或服務
- * **solution — chemistry** = 溶液
- * **solvable** = 可解決的／有解的
- * **unsolvable** = 無法解決的／無解的
- * **solver** = 解題者／求解器
- * **problem-solving** = 解決問題／問題解決能力
- * **solution-oriented** = 以尋找解決方法為導向的

A word such as **soluble** may look related, but in modern learner vocabulary it belongs primarily to the chemistry idea **「可溶解的」**, not to the ordinary **solve a problem** family. I would therefore keep it separate rather than artificially adding it here.

Master comparison

Form / sense	MCQ-ready Traditional Chinese meaning
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solve a problem	找到並實施有效方法, 使一個困難得到處理或消除
solve a puzzle/riddle	經過思考找出謎題或謎語的正確答案
solve an equation	透過數學步驟找出符合方程式條件的未知數值
solve for + variable	計算並找出指定未知數的值
solve a mystery	找出原本不明或神秘事件背後的真正解釋
solve a crime/case	找出犯案者及案件經過, 使案件得到偵破
solution — problem	能有效處理、減少或消除某個問題的方法／方案
solution — answer	經推理或計算後所得的正確解答
solution — business/technology	為處理特定需求或問題而提供的一套產品、服務或系統
solution — chemistry	一種物質均勻溶解於另一種物質後形成的混合物／溶液
solvable	存在可行方法或答案, 可以被成功解決的
unsolvable	沒有可行答案或在現有條件下無法成功解決的
solver	能找出問題、謎題或方程式答案的人、程式或工具
problem-solving	分析問題並找出有效解決方案的過程或能力
solution-oriented	面對問題時主要著眼於找出可行解決方法的

For your original sentence

> “Overall, that advertisement was persuasive because it ****solved a real problem****...”

The exact meaning is:

****solved = 產品真正提供了一個有效方法, 處理了使用者日常生活中實際存在的困難****

The ****real problem**** here is the noisy environment during commuting or travelling.

So:

> ****it solved a real problem****

= ****「它確實解決了一個現實生活中的實際問題。」****

The strength of ****solve**** here is important: it implies more than merely ****helping with**** the problem—it suggests that the headphones provided a genuinely effective answer to it.